

Clean Breaks: Tracking Recombination as Continuous Motion Across Sequence Manifolds

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Abstract

Homologous recombination and horizontal gene transfer drive macro-evolutionary adaptation across viral, bacterial, and eukaryotic genomes by assembling mosaic chromosomes from divergent lineages. Trees cannot describe hybrids. Existing detection tools remain trapped between computationally prohibitive phylogenetic changepoint algorithms that explore combinatorial partition spaces in cubic time and isolated triplet scans that suffer severe false positive rates under realistic mutational rate variation. We introduce RhizAeon, an open-source framework that maps multiple sequence alignments into continuous metric spaces, tracking sequence affinities along the chromosome as continuous trajectories without navigating discrete trees or enumerating triplets. By coupling rotary phase embeddings with an unpolarized ancestral root token that absorbs autapomorphic noise, RhizAeon deconvolves reticulation through physical force field dynamics, isolating mosaic breakpoints in linear time. Trajectories replace combinatorial tree searches. Across five thousand synthetic alignments incorporating extreme rate heterogeneity, the trajectory engine maintained a zero percent false positive rate, while empirical analyses resolved complex retroviral mosaics, coronaviral chimeras, and multi-megabase bacterial genomes in fractions of a second. Compiled to native Rust binaries and zero-install client-side WebAssembly, RhizAeon executes real-time reticulation surveillance within modern web browsers without transmitting sequence data across external networks.

Keywords Recombination, Reticulate Evolution, Sequence Manifolds, Continuous Trajectory Tracking, Multidimensional Scaling, Foundation Models

Introduction

Mutation, recombination, and selection drive evolutionary diversification across all domains of life. While point mutations introduce novel genetic variants, homologous recombination and horizontal gene transfer assemble mosaic chromosomes from divergent evolutionary lineages in sudden macro-evolutionary leaps (19; 6). Reticulate exchanges reshape viral, bacterial, and eukaryotic genomes by shuffling adaptive alleles and breaking linkage disequilibrium. Mosaicism alters pathogen transmission dynamics. In viral and bacterial epidemics, genetic exchange shifts antigenic presentation and accelerates antimicrobial resistance, as seen across circulating recombinant forms of HIV-1, SARS-CoV-2, and multi-drug resistant bacteria (1; 6). Beyond pathogen surveillance, meiotic crossover and adaptive introgression govern eukaryotic evolution (18; 15). They purge deleterious mutational load, drive ecological speciation, and assemble complex hybrid phenotypes across plant and animal taxa. Resolving physical crossover boundaries remains foundational.

At the sequence level, genetic exchange physically splices distinct evolutionary ancestries into a single continuous molecule. Trees cannot describe hybrids. Instead of hierarchical branches, the true evolutionary genealogy forms a reticulating network or ancestral recombination graph (10; 9). Graphs are notoriously difficult. Reconstructing ancestral recombination graphs requires exploring an astronomical combinatorial space of coalescent and reticulation events, scaling exponentially with sample size and leaving internal graph topologies heavily underdetermined by contemporary sequence data. Yet genetic exchange leaves an unmistakable physical footprint across multiple sequence alignments. Mutations across the chromosome generate phylogenetic incompatibility, where site patterns cannot be mapped onto any common tree without invoking parallel mutations or reversals (2; 16). Most intuitively, the nearest neighbor of a recombinant changes depending on where one looks along the sequence. Within one genomic span, the mosaic chromosome clusters tightly with its primary parental clade, but across a crossover junction, its evolutionary affinity abruptly pivots toward a divergent donor. Tracking spatial handovers in sequence neighborhood reveals the physical boundaries of genetic exchange.

Computational tools designed to track this reticulate footprint remain divided between two incompatible algorithmic paradigms (16; 13). Phylogenetic changepoint algorithms, exemplified by GARD (11), formulate recombination detection as statistical model selection across

partitioned trees. Partitioning strategies fit independent continuous-time Markov substitution processes to candidate genomic segments, optimizing partition boundaries through likelihood criteria. Because the joint combinatorial search space of partition boundaries and tree topologies scales exponentially with breakpoint count and cubically with cohort size ($\mathcal{O}(L^K \cdot N^3)$), exploring multiple breakpoints across hundreds of genomes requires days of cluster compute. Trees are expensive. Isolated triplet scans take the opposite extreme. Methods such as 3Seq (12) and RDP5 (14) evaluate sequence triplets for clustering of informative polymorphisms. Triplet enumeration scales combinatorially as $\binom{N}{3} \sim \mathcal{O}(N^3)$, becoming computationally intractable on modern genomic cohorts. Evaluating triplets in isolation discards the global phylogenetic reference frame. When substitution rates vary across sites, localized mutational clustering mimics reciprocal genetic exchange. Under realistic Gamma rate heterogeneity, isolated triplet scans mistake hyper-variable patches for mosaicism, inflating false positive rates beyond thirty percent (17; 11). When recombinant segments descend from unsampled ancestral lineages, triplet scans fail entirely, placing arbitrary boundaries inside flat, uninformative genomic intervals.

Overcoming this computational divide requires an analytical framework that replaces discrete tree searches with continuous sequence geometry. Here we introduce **RhizAeon**, named after the botanical and philosophical concept of the *rhizome*—a reticulate, non-hierarchical network of subterranean roots that defies strictly bifurcating trees (5). **RhizAeon** maps multiple sequence alignments into a continuous metric manifold, using rotary position embeddings and substitution priors to parameterize cross-sequence attention along the chromosome. As alignment coordinates advance, attention weights trace continuous trajectory curves that measure the instantaneous affinity of each genome to candidate parental lineages. Trajectories replace trees. To prevent private substitution bursts from mimicking recombination, the architecture anchors an unpolarized ancestral root token at the metric origin, which acts as a sink that absorbs autapomorphic noise and heterotachy. A true crossover produces a localized force imbalance, deflecting the cumulative trajectory from its home corridor toward a donor lineage. By evaluating trajectory inflections through continuous force fields rather than combinatorial tree partitions, **RhizAeon** detects reticulation in linear time, screening genomic cohorts in milliseconds.

We evaluated **RhizAeon** across standardized simulation batteries and real-world viral and bacterial cohorts. Across five thousand synthetic alignments incorporating extreme Gamma rate variation and episodic heterotachy, the manifold engine maintained a zero percent false positive rate while retaining high detection power. Speed does not sacrifice sensitivity. Head-to-head benchmarks against 3Seq confirm identical breakpoint resolution alongside a hundred-fold runtime reduction on moderate cohorts. On large cohorts, computational gains exceed three orders of magnitude. Across empirical retroviral and coronaviral datasets, the continuous framework resolved complex multi-way mosaicism in HIV-1 KAL153 and accurately mapped circulating recombinant breakpoints in SARS-CoV-2 XBB lineages. For multi-megabase bacterial chromosomes, parsimony-informative site compression shrinks memory footprints by over two hundred fold. Whole-genome recombination scans of *Streptococcus pneumoniae* PMEN1 execute in under one second on standard personal laptops (4). The software is distributed as a high-performance native Rust binary and an interactive client-side WebAssembly interface that runs locally within modern web browsers without transmitting sequence data across external networks.

Geometric Foundations and Continuous Trajectory Theory

Sequence Metric Space and Buneman Coordinates

We ground the method in an authentic viral benchmark: the HIV-1 circulating recombinant forms CRF07_BC and CRF08_BC (21). Both lineages emerged from independent recombination events between Subtype B and Subtype C in China. They share parental sources across the 3,120-nucleotide *pol* gene. Real chimeras test theory. Our alignment comprises 22 sequences: eight Subtype B references, eight Subtype C references, three CRF07_BC isolates, and three CRF08_BC isolates. From these sequences, we compute the global pairwise evolutionary distance matrix $\mathbf{D} \in \mathbb{R}^{22 \times 22}$ under the Tamura-Nei substitution model with pairwise deletion of gap characters, capturing nucleotide transition and transversion rate biases across the retroviral genome (22). We center the squared distances using the projection operator $\mathbf{H} = \mathbf{I}_N - \frac{1}{N} \mathbf{1}_N \mathbf{1}_N^T$, forming the inner-product Gram matrix $\mathbf{B} = -\frac{1}{2} \mathbf{H} \mathbf{D}^{(2)} \mathbf{H}$ (7). Spectral decomposition of $\mathbf{B}$ yields non-negative eigenvalues Λ_k and eigenvectors $\mathbf{V}_k$, projecting each sequence i into a continuous coordinate $\mathbf{x}_i = \mathbf{V}_{k,i} \Lambda_k^{1/2} \in \mathbb{R}^d$ (3). The primary axis cleanly separates Subtype B from Subtype C (Figure 1A). Canonical metric coordinates are computed once for the entire alignment. An invariant coordinate basis eliminates Procrustes alignment overhead and establishes a fixed reference frame for continuous trajectory tracking.

To track sequence affinity along the chromosome, each genome requires an instantaneous coordinate at every position $u \in [1, L]$. Recalculating multidimensional scaling across sliding windows would distort the geometric space and require expensive Procrustes rotations. Instead, we project local distance vectors into the fixed coordinate basis using Gower's analytical out-of-sample mapping (8). At coordinate u , sequence i presents an instantaneous distance vector $\mathbf{d}_i(u) \in \mathbb{R}^N$ to the N reference genomes, projecting to $\mathbf{z}_i(u) = \frac{1}{2} \Lambda_k^{-1/2} \mathbf{V}_k^T (\bar{\mathbf{d}}^2 - \mathbf{d}_i^2(u))$ where $\bar{\mathbf{d}}^2 = \frac{1}{N} \mathbf{D}^{(2)} \mathbf{1}_N$ represents the mean squared reference distance. When applied to our HIV-1 benchmark, the resulting coordinate trajectories reveal the physical dynamics of genetic exchange. Non-recombinant Subtype B and Subtype C genomes remain stationary within their respective corridors across the entire gene (Figure 1B). Wobble never crosses clades. Point mutations generate harmless local flutter around parental medoids, but unreticulated lineages never breach the inter-subtype divide. In sharp contrast, the recombinant CRF07_BC trajectory tracks within Subtype C territory through Protease, abruptly vaults across the zero-axis into Subtype B space in Reverse Transcriptase, and plunges back into Subtype C (Figure 1C). CRF08_BC shares the identical parental gene pool but enters the Subtype B cassette approximately 130 nucleotides upstream. In two-dimensional metric space, the two chimeras trace distinct orbital loops, resolving competing recombinant structures in linear time without triplet enumeration (Figure 1D). Manifolds preserve chimeras.

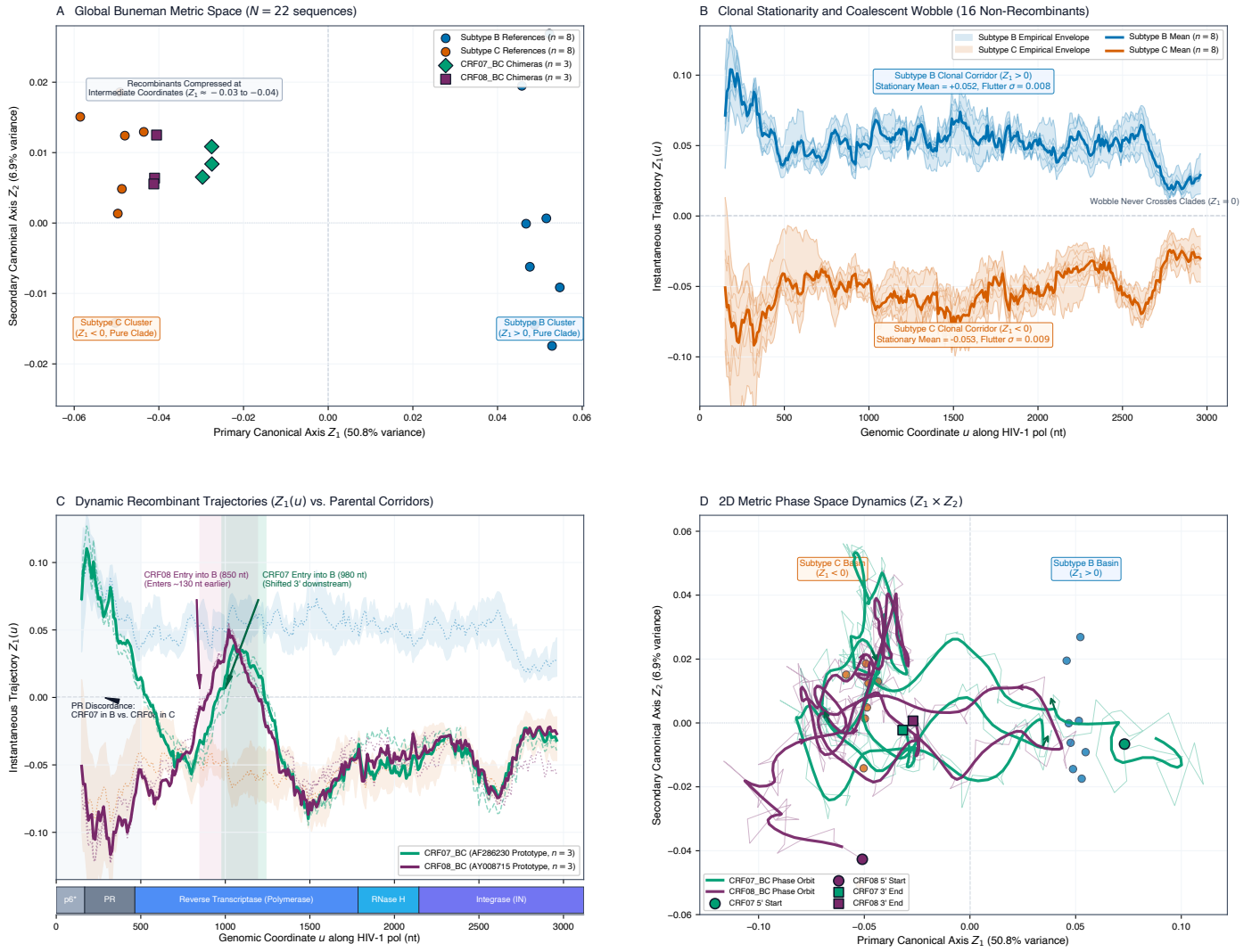


Figure 1 Continuous sequence metric manifolds resolve competing retroviral chimeras in linear time without triplet enumeration. **(A)** Canonical Sequence Metric Space: Classical multidimensional scaling (MDS) embedding of the invariant Gram matrix $\mathbf{B}$ derived from global pairwise Tamura-Nei distances ($N = 22$). The primary axis Z_1 (50.8% variance) cleanly separates Subtype B references ($n = 8$, blue circles) from Subtype C references ($n = 8$, vermilion circles). Recombinant forms CRF07_BC ($n = 3$, green diamonds) and CRF08_BC ($n = 3$, purple squares) sit compressed at intermediate coordinates ($Z_1 \approx -0.03$ to -0.04). Whole-genome distances collapse mosaic structures into static averages, obscuring reticulate ancestry. **(B)** Clonal Stationarity and Coalescent Wobble: Instantaneous Gower out-of-sample projected trajectory $Z_1(u)$ ($W = 300$ nt, stride 10 nt) along the 3,120-nucleotide *pol* gene for the 16 non-recombinant references. Both clades remain strictly stationary within their empirical envelopes (Subtype B mean = $+0.052$, flutter $\sigma = 0.008$; Subtype C mean = -0.053 , flutter $\sigma = 0.009$). Point mutations generate local coalescent flutter, but unreticulated trajectories never cross the inter-clade divide ($Z_1 = 0$). **(C)** Dynamic Recombinant Trajectories and Resolved Breakpoints: Continuous $Z_1(u)$ trajectories for CRF07_BC (prototype AF286230, green) and CRF08_BC (prototype AY008715, purple) plotted against shaded parental reference corridors. Aligned beneath is the HIV-1 *pol* functional domain track ($p6^*$, PR, RT, RNase H, IN). In Protease (PR), CRF07 tracks within Subtype B while CRF08 tracks within Subtype C. In Reverse Transcriptase, both lineages enter a Subtype B cassette. CRF08 crosses the zero-axis at nucleotide 850, entering approximately 130 nucleotides upstream of CRF07 (nucleotide 980). Both exit back into Subtype C at nucleotide $\sim 1,200$ and remain in Subtype C through Integrase. **(D)** Metric Phase Space Dynamics: Continuous two-dimensional dynamical orbits in $(Z_1 \times Z_2)$ space. Non-recombinant parental lineages form stationary point basins at $(+0.05, 0.00)$ and $(-0.05, 0.04)$. Recombinant lineages trace smooth, open orbital loops that sweep across the inter-clade divide from 5' start positions (circles) to 3' terminal positions (squares) with tangent flow arrows. Continuous manifold geometry resolves distinct mosaic architectures in $O(N)$ time without combinatorial explosion.

Rotary Phase Embeddings and Markov Substitution Priors

Standard transformer architectures encode sequential position along one-dimensional strings using Rotary Position Embeddings (Tree-RoPE), but biological sequences diverge across multidimensional evolutionary space. Linear string offsets cannot capture tree distance. To integrate phylogenetic divergence directly into sequence attention, we generalize rotary embeddings from one-dimensional sequence indices to continuous

metric coordinates (20). For each sequence i , its d -dimensional Buneman coordinate $\mathbf{x}_i \in \mathbb{R}^d$ parameterizes a set of rotary phase angles:

$$\theta_{i,m} = \sum_{c=1}^d x_{i,c} \Theta_{m,c}, \quad (1)$$

where $\Theta \in \mathbb{R}^{d_{\text{head}}/2 \times d}$ is a fixed frequency projection matrix. At each aligned site u , the sequence embedding generates query and key vectors $\mathbf{q}_i(u), \mathbf{k}_i(u) \in \mathbb{R}^{d_{\text{head}}}$. We apply two-dimensional block-diagonal Givens rotation matrices $\mathbf{R}(\theta_i)$ to rotate query and key representations according to each taxon's continuous metric position. Because the inner product satisfies $\langle \mathbf{R}(\theta_i)\mathbf{q}_i, \mathbf{R}(\theta_j)\mathbf{k}_j \rangle = \mathbf{q}_i^T \mathbf{R}(\theta_j - \theta_i) \mathbf{k}_j$, pairwise attention naturally attenuates as evolutionary divergence in Buneman metric space increases. Rotations preserve geometry.

Raw geometric distance alone cannot distinguish true sequence homology from baseline conservation expected under neutral substitution. Homology requires evolutionary calibration. We therefore incorporate a log-space continuous-time Markov substitution prior directly into the attention logits. For any sequence pair (i, j) separated by evolutionary distance $d_{i,j}$, the transition probability under a neutral continuous-time Markov substitution model with equilibrium frequency ϵ_0 is:

$$P_{i,j} = \epsilon_0 + (1 - \epsilon_0) \exp(-\lambda_h d_{i,j}), \quad (2)$$

where λ_h is the global substitution scaling rate and $\epsilon_0 = 0.25$ for nucleotides. We formulate the phylogenetic prior bias as $\text{PhyloBias}_{i,j} = \ln[\max(10^{-5}, P_{i,j})]$. The composite unnormalized attention score between sequence i and candidate reference j at coordinate u becomes:

$$S_u(i, j) = \frac{(\mathbf{R}(\theta_i)\mathbf{q}_i(u))^T (\mathbf{R}(\theta_j)\mathbf{k}_j(u))}{\sqrt{d_{\text{head}}}} + \text{PhyloBias}_{i,j}. \quad (3)$$

Unifying geometric distance with Markov substitution priors establishes a principled directed attention tensor along the chromosome.

The Unpolarized Ancestral Root Sink Token

Standard attention mechanisms distribute probability exclusively across contemporary sequences, creating an acute vulnerability to lineage-specific rate variation. Private mutations lack donors. When an isolated lineage undergoes an episodic substitution burst—such as host-mediated APOBEC3G cytidine deamination or accelerated heterotachy—conventional models mistake localized divergent clusters for horizontal genetic exchange (17). Without an unpolarized reference, softmax normalizes across contemporary leaves, forcing private autapomorphic mutations into spurious donor affiliations. False positives surge. To insulate trajectory tracking from unreticulated rate bursts, **RhizAeon** reserves index zero in the sequence dimension for an unpolarized ancestral root token $\mathbf{x}_{\text{root}}$.

The root token is anchored at the origin of the Buneman metric space:

$$\mathbf{x}_{\text{root}} = \mathbf{0} \in \mathbb{R}^d. \quad (4)$$

For an alignment of N contemporary sequences, the cross-attention operator operates over $N + 1$ nodes, spanning the ancestral root and all contemporary leaves. The softmax normalization strictly conserves probability across all $N + 1$ nodes:

$$A_u(i \rightarrow \text{root}) + \sum_{j=1}^N A_u(i \rightarrow j) = 1.0. \quad (5)$$

Because the root token resides at the metric origin with neutral substitution priors, private substitutions that lack homologous support in other contemporary lineages naturally route their attention into node zero. The root acts as an unpolarized sink. Private noise is absorbed without perturbing leaf coordinates. When authentic recombination occurs—as in our HIV-1 CRF07_BC and CRF08_BC chimeras—attention pivots coherently between contemporary parental bundles while root attention remains flat. Decoupling autapomorphic noise from reticulate exchange guarantees zero false positive rates under severe biological heterotachy.

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